
EduRAG: Verified and Stateless Cognitive Load-Aware Adaptation for Retrieval-Augmented Generation

RESEARCH PROPOSAL

AUTHOR(S):

Ahmad Raza (SOFTF22BSR04)

Abdullah Khan (SOFTF22BSR28)

DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
THAL UNIVERSITY BHAKKAR

Abstract

Educational Retrieval-Augmented Generation (RAG) systems face a fundamental, unaddressed tension: knowledge grounding mechanisms assume output should approximate source form, but pedagogical effectiveness demands cognitive-level transformation — simplification, analogy, and instantiation. This research presents **EduRAG**, a lightweight offline-first RAG architecture addressing this *Adaptation-Grounding Tension* through three primary contributions and two application demonstrations:

Primary Contributions:

1. **CLAG (Cognitive Load-Aware Generation)**: A stateless cognitive adaptation framework that estimates student expertise solely from query linguistics i.e., Bloom’s level, specificity, and complexity without persistent user models, producing a 5-dimensional adaptation vector grounded in Cognitive Load Theory (Sweller, 1988) and the Expertise Reversal Effect (Kalyuga, 2007). CLAG challenges the prevailing “Memory Dogma” that adaptation requires persistent learner models.
2. **AGR (Adaptation-Grounding Resolution)**: The first derivation-aware verification framework that defines transformation rules (Substitution, Instantiation, Simplification, Analogy, Implication) for validating pedagogically adapted content, resolving the tension between adaptation and grounding by introducing the DERIVED verification category absent from all prior verification frameworks (FactScore, VeriScore, VeriCite).
3. **ERSA (Educational RAG Scalability Analysis)**: A systematic empirical characterization of retrieval scaling behavior across 44 experimental conditions, introducing the Retrieval Scaling Coefficient (RSC) via AIC/BIC model selection, the Educational Viability Threshold (EVT), and the Query-Type Sensitivity Index (QTSI).

Application Demonstrations:

1. **PCO (Pedagogical Context Ordering)**: A zero-overhead post-retrieval reordering mechanism validated through user studies ($N = 24\text{--}30$).
2. **CAQ (CLAG-Aligned Quiz Generation)**: An adaptive assessment system leveraging CLAG-computed state for quiz difficulty calibration without IRT pre-testing.

For knowledge grounding, we adopt the Structure-Aware RAG paradigm established by **KiRAG** (Fang et al., 2025), **RECOMP** (Xu et al., 2023), and **LongCite** (Zhang et al., 2024). We target offline deployment on consumer hardware using **LLaMA 3.1 8B** (4-bit quantized), demonstrating that principled educational AI is achievable without cloud infrastructure.

1. Research Problem Context

1.1 The Adaptation-Grounding Tension

Educational AI systems bear heightened responsibility for factual accuracy, erroneous information delivered with pedagogical authority can cement misconceptions (Kasneci et al., 2023). Structure-Aware RAG systems (KiRAG, Fang et al., 2025; RECOMP, Xu et al., 2023; LongCite, Zhang et al., 2024) have largely solved the extraction and citation problem: how to ground LLM generation in structured knowledge units from retrieved documents.

However, educational use imposes a requirement absent from general-purpose RAG: **the pedagogically correct response must differ from the source form**. A student asking “what is binary search?” should not receive “ $O(\log n)$ worst-case comparison-based search in a sorted array with lower bound from the adversary argument.” They should receive an adapted explanation with analogies, concrete instantiations, and simplified vocabulary.

This creates a fundamental conflict. Standard verification mechanisms i.e., BERTScore (Zhang et al., 2020), NLI entailment (Honovich et al., 2022), FActScore (Min et al., 2023), VeriScore (Song et al., 2024) assume output should approximate source form. When pedagogical adaptation transforms “ $O(\log n)$ comparisons” into “about 20 comparisons for a million elements” (via $\log_2(10^6) \approx 20$), these methods produce false positives: the adaptation is semantically valid but lexically divergent.

We formalize this as the **Adaptation-Grounding Tension**: the irreconcilable conflict between (a) grounding constraints that penalize deviation from source, and (b) pedagogical constraints that *require* deviation from source.

1.2 The Stateless Adaptation Challenge

Existing adaptive learning systems i.e., Intelligent Tutoring Systems (VanLehn, 2011), Bayesian Knowledge Tracing (Corbett & Anderson, 1995) universally assume persistent learner models with mastery tracking, error history, and session analytics. For stateless RAG systems particularly offline deployments on consumer hardware where no login, database, or session tracking exist, these assumptions fail entirely.

Definition (Stateless Cognitive Adaptation). Given a single query q and retrieved context D , estimate the student’s cognitive level solely from (q, D) and produce a response adapted to that level without any session history, persistent user model, mastery score, error log, or prior interaction data. The prevailing orthodoxy (**Memory Dogma**) holds that adaptation requires remembering the learner.

1.3 The Scalability Blindspot

Educational RAG systems must accommodate growing knowledge bases as curricula expand. No systematic study has characterized how retrieval performance scales with corpus growth in curriculum-structured educational corpora. Critical questions remain unanswered:

- Does $P@k$ degrade linearly, logarithmically, or exhibit threshold effects?
- Which architectural factors (chunking, embedding model, retrieval method) most significantly moderate degradation?
- At what corpus size does quality become unacceptable for educational deployment?

1.4 Knowledge Grounding Infrastructure

The knowledge grounding pipeline i.e., knowledge unit extraction from retrieved documents, constrained generation via soft-constraint prompting, and generation-time citation tagging, has been independently developed by multiple groups:

Paper	Year/Venue	Contribution
KiRAG (Fang et al.)	2025, arXiv:2502.18397	Knowledge triple extraction + iterative retrieval
RECOMP (Xu et al.)	2023, ICLR 2024	T5-based extractive/abstractive context compression
LongCite (Zhang et al.)	2024, ACL 2025 Findings	SFT-trained generation-time citation
Saxena et al.	2025, NeurIPS Workshop	G-Cite vs. P-Cite paradigm characterization
FAIR-RAG (Asl et al.)	2025	Soft-constraint prompting for faithful generation

We adopt this paradigm as System Architecture, not as a contribution, and cite these works as our foundation.

2. Research Questions and Hypotheses

2.1 Primary Research Questions

2.1.1 Stateless Cognitive Adaptation — CLAG *Can query-derived linguistic signals (Bloom’s level, specificity, complexity) provide sufficient information for principled cognitive load adaptation in stateless educational RAG, achieving $\geq 80\%$ agreement with expert cognitive-level assessment?*

Sub-Questions:

- What is the accuracy of pattern-based Bloom’s level detection on educational CS queries? (Target: $> 80\%$, macro-F1 > 0.75)
- Does CLAG-adapted generation improve perceived appropriateness over fixed-level generation? (Target: $p < 0.05$, Cohen’s $d > 0.5$)
- Which component of the QD formula (B, S, L) contributes most to adaptation quality? (Ablation)

2.1.2 Adaptation-Grounding Resolution — AGR *Can derivation-aware verification distinguish valid pedagogical transformations from hallucinations, reducing false positive rates by $\geq 40\%$ over surface-matching baselines while maintaining $\geq 95\%$ true positive rate?*

Sub-Questions:

- What is the distribution of transformation types (Substitution, Instantiation, Simplification, Analogy, Implication) in educational RAG responses? (Descriptive)
- What is the false positive rate of standard verification (BERTScore, NLI) on pedagogically adapted content? (Baseline measurement)
- Does AGR’s derivation rule engine significantly reduce false positives? (Comparative)

2.1.3 Retrieval Scaling — ERSA *What functional form best characterizes retrieval precision scaling as educational corpora scale from 2 to 8 semesters, and at what corpus size does performance cross the educational viability threshold?*

Sub-Questions:

- Among Linear, Log-linear, Power-law, and Exponential models, which best fits the $P@k$ -corpus size relationship? (AIC/BIC model selection)
- Do different query types exhibit significantly different scaling behavior? (QTSI analysis)
- What is the EVT for typical educational RAG configurations? (EVT estimation)

2.2 Secondary Research Questions

- **Ordering:** Does pedagogically-motivated context ordering improve comprehension compared to similarity-ranked retrieval?
- **Assessment:** Can CLAG-computed cognitive state drive adaptive quiz generation without IRT calibration?

2.3 Testable Hypotheses

ID	Hypothesis	Operationalization
H1a	CLAG Bloom’s detection achieves $> 80\%$ accuracy	300 queries, 2 CS educator annotators, Cohen’s $\kappa > 0.75$
H1b	CLAG adaptation outperforms fixed-level generation	Between-subjects study ($N = 60$), one-way ANOVA, $p < 0.05$, $d > 0.5$
H2a	Standard verification FPR $> 15\%$ on adapted content	Baseline measurement on 300 transformation pairs
H2b	AGR reduces FPR by $\geq 40\%$ vs. surface-matching	Comparative: AGR-Full vs. BM25/BERTScore/NLI
H2c	AGR maintains TPR $\geq 95\%$	Same benchmark
H3a	Log-linear best fits $P@k = f(N)$ in $> 60\%$ of conditions	AIC/BIC model selection across 44 conditions
H3b	Semantic chunking exhibits 20–40% lower RSC than fixed-size	Pairwise at $N > 10,000$ chunks
H3c	Hybrid retrieval exhibits 30–50% lower RSC than dense-only	Controlled comparison across scales
H4	PCO improves coherence by > 0.8 points (5-point scale)	Within-subjects, $N = 24\text{--}30$
H5	CAQ quizzes achieve $> 85\%$ Bloom’s alignment	Automated + expert validation

3. Technical Contributions

3.1 CLAG: Cognitive Load-Aware Generation (Primary 1)

3.1.1 Motivation The query itself reveals student cognitive level. A student asking “What is a binary search tree?” (Remember) operates at a different cognitive level than one asking “Why is BST $O(n)$ with sorted input?” (Analyze). All adaptive learning systems assume the **Memory Dogma**: “You cannot adapt without remembering.” CLAG’s hypothesis is that a single query’s linguistic fingerprint contains sufficient signal for meaningful cognitive load estimation — a testable, falsifiable hypothesis that challenges the prevailing paradigm.

3.1.2 Theoretical Grounding

Theory	Citation	Connection to CLAG
Cognitive Load Theory	Sweller (1988); Sweller et al. (1998)	CLT defines intrinsic, extraneous, germane load. CLAG’s adaptation vector controls extraneous load via vocabulary, density, scaffolding
Bloom’s Revised Taxonomy	Anderson & Krathwohl (2001)	$B(q)$ maps queries to 6 cognitive levels. Higher Bloom’s $\rightarrow$ higher intrinsic load capacity
Expertise Reversal Effect	Kalyuga (2007); Kalyuga et al. (2003)	Over-scaffolding harms experts. CLAG reduces scaffolding for high- QD queries
Zone of Proximal Development	Vygotsky (1978)	Content slightly above current query level $\rightarrow$ optimal learning
Desirable Difficulties	Bjork (1994); Bjork & Bjork (2011)	Avoids making content too easy; high- QD gets less hand-holding

3.1.3 Mathematical Formulation

Query Demand (QD) — Optimal Formula:

$$QD(q) = B(q) + \text{clamp}\left(\beta_S \cdot (S(q) - \frac{1}{2}) + \beta_L \cdot (L(q) - \frac{1}{2}), -\delta_q, +\delta_q\right)$$

Where:

- $B(q) = \frac{\text{BloomLevel}(q)-1}{5}$: Normalized Bloom’s level $([0, 1])$ — **primary signal** (effective weight 1.0)
- $S(q)$: Query specificity $([0, 1])$ — bounded auxiliary correction
- $L(q)$: Linguistic complexity $([0, 1])$ — bounded auxiliary correction
- $\beta_S = 0.3, \beta_L = 0.2$: Sensitivity parameters (preserving 3:2 S/L ratio)

- $\delta_q = 0.10$: Maximum allowed deviation from Bloom estimate (0.5 Bloom levels)
- Centering at $S - \frac{1}{2}$, $L - \frac{1}{2}$ ensures zero average correction for typical queries

Design Rationale: The original linear formula $QD = w_1B + w_2S + w_3L$ assigns only 50% weight to Bloom’s level, allowing S and L to cause up to 2.5 Bloom levels of worst-case error. The Optimal formula uses B as the base estimate and clamps auxiliary corrections to $\pm\delta_q$, guaranteeing $|QD - B| \leq 0.10$ always. This dissolves the critical correlation requirement ($\rho^* = 0.676$) that the linear formula faces — the Optimal formula never degrades more than $\delta_q^2 = 0.01$ in MSE above Bloom-only, regardless of whether S and L correlate with cognitive level.

Bloom’s Level Detection (Pattern-Based):

Level	Patterns	Example
1: Remember	“what is”, “define”, “list”	“What is a stack?”
2: Understand	“explain”, “describe”, “how does... work”	“How does recursion work?”
3: Apply	“implement”, “write code”, “how to”	“How to implement quicksort?”
4: Analyze	“compare”, “why does”, “difference”	“Compare arrays vs linked lists”
5: Evaluate	“which is better”, “pros and cons”	“Should I use SQL or NoSQL?”
6: Create	“design”, “propose”, “improve”	“Design a cache system”

Content Complexity (CC):

$$CC(D) = 0.4 \cdot \text{ConceptDensity}(D) + 0.3 \cdot \text{Readability}(D) + 0.3 \cdot \text{AbstractionLevel}(D)$$

Where D = retrieved chunks.

Adaptation Level:

$$\text{AdaptationLevel}(q, D) = \alpha \cdot QD(q) + (1 - \alpha) \cdot \text{clamp}(CC(D), QD(q) - \delta, QD(q) + \delta)$$

where $\alpha = 0.7$ (QD primacy) and $\delta = 0.2$ (clamp radius).

Properties:

- Bloom’s level (B) is the primary signal with effective weight 1.0 in QD; S and L contribute bounded corrections only
- QD primacy: $\alpha = 0.7$ ensures QD drives $\geq 70\%$ of AL (100% when CC is outside the clamp window)
- **Guaranteed ZPD Targeting:** Under perfect Bloom detection, $|AL - \theta| \leq \delta_q + (1 - \alpha)\delta = 0.16$ (within 1 Bloom level, always). Under typical detection ($p = 0.833$): $|AL - \theta| \leq 0.36$ for the 16.7% of misdetected queries

- **AdaptationLevel** $\in [0, 1]$: 0 = maximum simplification, 1 = maximum technicality
- **No Critical Correlation Threshold:** Unlike linear QD formulas which require $\rho_{S,L} > 0.676$ to outperform Bloom-only, the Optimal formula works for all $\rho \geq 0$.

CLT Grounding:

- **Low QD:** high scaffolding, simple vocabulary $\rightarrow$ reduces extraneous load (Sweller, 1988)
- **High QD:** minimal scaffolding, technical vocabulary $\rightarrow$ avoids Expertise Reversal Effect (Kalyuga, 2007)
- **CC clamping:** targets Zone of Proximal Development (Vygotsky, 1978)

Adaptation Vector $\mathbf{A} = (v, d, e, s, l)$:

Dimension	Low Adaptation (Novice)	High Adaptation (Expert)
Vocabulary (v)	Simple language, define terms	Technical terminology
Density (d)	1 concept per response	3–4 concepts, edge cases
Examples (e)	3 concrete, familiar examples	1 abstract example
Scaffolding (s)	Step-by-step, prerequisites	Direct, efficient
Length (l)	Short, chunked	Comprehensive, dense

3.1.4 Differentiation from Prior Art

System	What It Does	Why CLAG Is Different
FEEG (Samuel et al., 2025)	Bloom’s classification to <i>evaluate</i> LLM quality	CLAG uses Bloom’s as <i>input signal</i> for <i>generation adaptation</i> via 5D adaptation vector
LQR-HyRAG (KR et al., 2025)	Query routing to retrieval strategies	Acts on retrieval, not generation output
AdaptLearn AI (Gupta et al., 2025)	Bidirectional personalization via user data	Stateful. Requires persistent user profiles
Adaptive-RAG (Jeong et al., 2024)	Complexity routing (no-retrieval/single/multi)	Routes retrieval, does not adapt response characteristics
Standard ITS (BKT, ECD)	Persistent mastery/error tracking	Requires session history; CLAG requires zero history

3.1.5 Experiment Design: CLAG Evaluation

Experiment C1: Bloom’s Detection Accuracy

- **Data:** 300 CS education queries (50 per Bloom’s level $\times$ 6 levels)
- **Annotators:** 2 CS educators, Cohen’s κ (target: > 0.75)

- **Conditions:** Random, Majority-class, CLAG-Patterns, CLAG-Full ($B + S + L$), Oracle-LLM (GPT-4)
- **Metrics:** Per-level accuracy, macro-F1, confusion matrix
- **Target:** CLAG-Full $> 80\%$ accuracy, macro-F1 > 0.75

Experiment C2: User Study for Adaptation Quality

- **N:** 60 CS students (between-subjects, 20 per condition)
- **Conditions:** Fixed-Beginner, Fixed-Advanced, CLAG-Adaptive
- **Task:** 10 CS questions across Bloom’s levels 1–5, pre/post comprehension test
- **Metrics:**
 - Appropriateness (Likert 1–5)
 - Difficulty match (Likert 1–5)
 - Comprehension gain: $\frac{\text{post-pre}}{\text{max-pre}}$
 - Engagement (Likert 1–5)
- **Analysis:** One-way ANOVA per metric, Bonferroni correction, effect sizes (Cohen’s d)
- **Target:** CLAG $>$ Fixed conditions at $p < 0.05$, $d > 0.5$

Experiment C3: Ablation

- **Conditions:** No-CLAG (fixed medium), Bloom-Only, Bloom+Specificity, Full-CLAG
- **Metrics:** Same as C2
- **Purpose:** Quantify marginal contribution of each QD component

Statistical Analysis:

- One-way ANOVA with Bonferroni correction
- Effect sizes: Cohen’s d (target: > 0.5)
- Inter-annotator agreement: Cohen’s κ (target: > 0.75)

3.2 AGR: Adaptation-Grounding Resolution (Primary 2)

3.2.1 Motivation The Adaptation-Grounding Tension creates a fundamental verification problem: standard verification penalizes valid pedagogical transformations. Source says “ $O(\log n)$ comparisons.” Adapted response says “about 20 comparisons for a million elements.” This is factually correct ($\log_2(10^6) \approx 20$) but will be flagged as:

- Low BERTScore (different surface form)
- NLI: NEUTRAL or CONTRADICTION (no training data for mathematical instantiation)
- FActScore: “20 comparisons” not found as atomic fact in source

No existing verification framework addresses this problem. The DERIVED verification category does not exist in any prior system.

3.2.2 Differentiation from Prior Art

Paper	Year	What They Do
FactScore (Min et al., 2023)	EMNLP 2023	Atomic fact decomposition + source verification
VeriScore (Song et al., 2024)	2024	Verifiable claim extraction + search
VeriCite (Qian et al., 2025)	ACM 2025	Citation verification in RAG
FactLens (Mitra et al., 2025)	ACL 2025	Benchmarks fact verification
GraphCheck (2025)	—	Graph-based post-gen fact checking

3.2.3 Mathematical Formulation

Verification Function:

$$\text{Verify}(c, \mathcal{KU}) = \begin{cases} \text{GROUNDED} & \text{if } \exists ku \in \mathcal{KU} : \text{match}(c, ku) \\ \text{DERIVED} & \text{if } \exists ku_1, \dots, ku_j \in \mathcal{KU}, \rho \in \mathcal{R} : \text{derives}(c, ku_1, \dots, ku_j, \rho) \\ \text{GENERAL} & \text{if } c \in \mathcal{G} \\ \text{UNVERIFIED} & \text{otherwise} \end{cases}$$

Category	Definition	Example	Action
GROUNDED	Direct entity/fact match with $\mathcal{KU}$	“Binary search uses divide and conquer” (source says same)	Accept
DERIVED	Valid transformation via derivation rule	“ $O(\log n)$ ” $\rightarrow$ “about 20 comparisons for 1M elements”	Accept with derivation trace
GENERAL	Common CS/general knowledge not in $\mathcal{KU}$	“Computers use binary”	Accept (no citation)
UNVERIFIED	Specific claim with no grounding or derivation	“Invented by Dijkstra” (not in $\mathcal{KU}$)	Flag

Derivation Rules $\mathcal{R}$:

Rule	Schema	Example
Substitution	Replace formal notation with equivalent informal expression	$O(\log n) \rightarrow$ “grows very slowly”
Instantiation	Plug concrete values into abstract formula	$\log_2(N)$ for $N = 10^6 \rightarrow$ “about 20”

Rule	Schema	Example
Simplification	Remove qualifiers while preserving core truth	“amortized expected $O(1)$ ” $\rightarrow$ “effectively constant time”
Analogy	Map to familiar domain preserving structure	“Stack works like a pile of plates”
Implication	Derive logical consequence from ≥ 1 KUs	“ A is $O(n \log n)$ ” + “ B is $O(n^2)$ ” $\rightarrow$ “ A is faster for large inputs”

Implementation:

- Substitution/Instantiation: Pattern matching + symbolic math (SymPy). Deterministic.
- Simplification/Analogy: Prompted LLM classification (confidence threshold = 0.7).
- Implication: Lightweight NLI (DeBERTa, entailment threshold = 0.8).
- Total overhead: < 50ms per response.

3.2.4 Experiment Design: AGR Evaluation

Experiment A1: Benchmark Construction

- **Data:** 300 (source, transformation) pairs
- **Types:** ~43 each of Substitution, Instantiation, Simplification, Analogy, Implication, Composite, Hallucination
- **Labels:** GROUNDED, DERIVED, GENERAL, UNVERIFIED
- **Annotators:** 2 CS educators + 1 NLP researcher, Cohen’s κ (target: > 0.75)

Experiment A2: Method Comparison

Condition	Description
Surface-Match	BM25-based lexical matching
Semantic-Sim	BERTScore-based semantic similarity
NLI-Verify	DeBERTa NLI entailment
AGR-NoRules	Entity matching only (no derivation rules)
AGR-Full	Full derivation-aware verification

Metrics: Precision, Recall, F1 per verification category, FPR, TPR, Overall Accuracy - **Targets:** AGR-Full FPR < 10%, TPR > 95%, F1(DERIVED) > 0.85, Accuracy > 90%

Experiment A3: False Positive Reduction

- **Data:** 100 CLAG-adapted responses with known transformations
- **Metric:** FP reduction % vs. Surface-Match and NLI baselines
- **Target:** > 40% FP reduction

Experiment A4: Integration Test

Condition	Description
No-Adaptation	Standard RAG, no CLAG
Adapt-NoVerify	CLAG adaptation, no verification
Adapt-Surface	CLAG + surface-level verification
Adapt-AGR	CLAG + AGR (full system)

Metrics: End-to-end hallucination rate, citation accuracy, user trust (Likert 1–5)

Statistical Analysis:

- McNemar’s test for accuracy comparisons
- Paired t-test for FPR comparisons
- Inter-annotator agreement: Cohen’s κ (target: > 0.75)

3.3 ERSA: Educational RAG Scalability Analysis (Primary 3)

3.3.1 Motivation No prior work systematically characterizes retrieval behavior in educational RAG under corpus scaling. ERSA fills this gap with:

1. **Empirical characterization** of scaling curves via model selection (not assumed functional form)
2. **Moderator analysis** identifying which factors mitigate degradation
3. **Predictive thresholds** via the Educational Viability Threshold (EVT)

We do not assert any functional form *a priori*. We fit multiple candidate models and select via information-theoretic criteria.

3.3.2 Mathematical Formulation Candidate Models:

Model	Functional Form	Parameters
Linear	$P@k(N) = \alpha - \beta N$	α, β
Log-linear	$P@k(N) = \alpha - \beta \log_2(N)$	α, β
Power-law	$P@k(N) = \alpha \cdot N^{-\beta}$	α, β
Exponential	$P@k(N) = \alpha \cdot e^{-\beta N}$	α, β

Model Selection: AIC and BIC across all 44 conditions. Report:

- Which model wins in what % of conditions
- $RSC = \beta$ of the winning model per condition
- RSC can be negative (improvement with scale)

Retrieval Scaling Coefficient (RSC):

$RSC = \beta$ of the best-fitting model, enabling cross-configuration comparison. Low $|RSC|$ = scalability-robust; High $|RSC|$ = scalability-fragile.

Educational Viability Threshold (EVT):

$$N_{\text{EVT}} = \min\{N : P@k(N) < 0.70 \text{ or } \text{HallucinationRate}(N) > 0.15\}$$

Predicts maximum viable corpus size for a given configuration.

Query-Type Sensitivity Index (QTSI):

$$\text{QTSI}_{\text{type}} = \frac{\text{RSC}_{\text{type}}}{\text{RSC}_{\text{overall}}}$$

QTSI > 1 = query type more vulnerable to scaling than average; QTSI < 1 = robust.

3.3.3 Experiment Design: ERSA**Independent Variables (Fractional Factorial Design):**

Factor	Levels	Rationale
Corpus Size	2, 4, 6, 8 semesters (~3.5k, 7k, 10.5k, 14k chunks)	Primary variable
Chunking Strategy	Fixed-512, Fixed-1024, Semantic, Hierarchical	Information granularity
Embedding Model	all-MiniLM-L6-v2, BGE-base-en, E5-base-v2	Representation quality
Retrieval Method	Dense-only, BM25-only, Hybrid (RRF)	Paradigm comparison

Experimental Conditions:

Full factorial: $4 \times 4 \times 3 \times 3 = 144$ conditions (infeasible).

Fractional factorial design (44 conditions):

- Baseline sweep: All corpus sizes $\times$ Fixed-512 $\times$ MiniLM $\times$ Dense (4)
- Chunking ablation: All corpus sizes $\times$ All chunking $\times$ MiniLM $\times$ Dense (16)
- Embedding ablation: All corpus sizes $\times$ Fixed-512 $\times$ All embeddings $\times$ Dense (12)
- Retrieval ablation: All corpus sizes $\times$ Fixed-512 $\times$ MiniLM $\times$ All methods (12)

Dependent Variables:

Category	Metrics
Retrieval Quality	P@5, P@10, R@10, MRR, NDCG@10
Educational Quality	Answer Accuracy, Hallucination Rate, Citation Accuracy
Efficiency	Retrieval Latency (ms), Index Memory (MB)

Query Stratification:

Query Type	Example	Expected Scaling Sensitivity
Factual/Definition	“What is quicksort’s complexity?”	HIGH (precise retrieval needed)
Procedural	“How to implement merge sort?”	MEDIUM
Conceptual	“Why is hashing $O(1)$?”	MEDIUM
Comparative	“Compare BFS vs DFS”	LOW (partial context sufficient)
Application	“Design an LRU cache”	LOW (reasoning-heavy)

Evaluation Protocol:

- 250 queries per condition (50 queries $\times$ 5 types)
- 2 annotators per query, Cohen’s $\kappa > 0.70$
- Total: $44 \times 250 = \mathbf{11,000}$ evaluations

Statistical Analysis:

1. Fit 4 candidate models per condition via OLS
2. Select best model per condition via AIC/BIC
3. Report model-win table
4. Define $RSC = \beta$ of best-fitting model
5. Compute EVT per configuration
6. Compute QTSI per query type
7. 2-way ANOVA: chunking $\times$ corpus size, embedding $\times$ retrieval, query-type $\times$ corpus size
8. Effect sizes: Cohen’s d , η^2 . Bonferroni correction for multiple comparisons.

3.4 Application: Pedagogical Context Ordering (PCO)

3.4.1 Motivation Standard RAG returns chunks ranked by similarity. Educational effectiveness requires pedagogical sequencing: definitions before procedures, fundamentals before applications.

3.4.2 Algorithm

PCO Classification Rules

1. FOUNDATION (ordered first)

Signals: "is defined as", "refers to", "is a type of"

Content: Definitions, basic concepts

2. CORE (ordered second)

Signals: "algorithm", "procedure", "implementation"

Content: How things work, procedures

3. ADVANCED (ordered third)

Signals: "optimization", "tradeoff", "compared to"

Content: Analysis, comparisons, edge cases

Final Order: Foundation $\rightarrow$ Core $\rightarrow$ Advanced

Within category: Preserve similarity ranking

Complexity: $O(k)$ where k = chunks. Adds ~ 0.1 ms latency.

Theoretical Grounding:

- Scaffolding Theory (Wood, Bruner & Ross, 1976)
- Cognitive Load Theory: Foundational knowledge reduces intrinsic load
- Advance Organizers (Ausubel, 1960)

3.4.3 Experiment Design: PCO User Study

Design: Within-subjects, 3 conditions

Participants: $N = 24$ – 30 CS students

Condition	Description
A: Similarity-ranked	Standard RAG ordering (baseline)
B: Random	Shuffled chunks (control)
C: PCO	Pedagogical ordering (treatment)

Procedure:

1. Each participant sees 6 explanations (2 per condition)
2. Topics counterbalanced via Latin square
3. After each explanation: Rate coherence, clarity, completeness (Likert 1–5)
4. Comprehension quiz (3 questions per topic)

Metrics: Coherence (Likert 1–5), Clarity (Likert 1–5), Completeness (Likert 1–5), Quiz accuracy (%)

Statistical Analysis: Repeated-measures ANOVA with Bonferroni correction, effect size via partial η^2

3.5 Application: CLAG-Aligned Quiz Generation (CAQ)

3.5.1 Motivation Traditional adaptive assessment requires IRT calibration with extensive pre-testing to estimate item difficulty parameters, creating a cold-start problem for new content. CAQ leverages CLAG’s query demand score for quiz adaptation without pre-testing.

3.5.2 CLAG State to Quiz Configuration Given CLAG state $S = (B, L, T)$ where B = Bloom’s level, L = load ratio, T = complexity tier:

$$n_{\text{questions}} = \begin{cases} 3 & \text{if } L > 0.7 \\ 5 & \text{if } 0.4 \leq L \leq 0.7 \\ 7 & \text{if } L < 0.4 \end{cases}$$

$$B_{\text{target}} \in [\max(1, B - 1), B]$$

Question Formats by Tier:

Tier	Formats	Scaffolding
Beginner	Fill-blank, True/False	Hints enabled, show source
Intermediate	Short-answer, Fill-blank	Optional hints
Advanced	Short-answer, Explain	No hints

3.5.3 Experiment Design: CAQ Evaluation

Experiment 1: Bloom’s Alignment

- Dataset: 200 generated questions, Evaluators: 2 educators label actual Bloom’s level
- Metric: Agreement between target and actual (target: > 85%)

Experiment 2: Difficulty Appropriateness

- Participants: 20 CS students
- Task: Interact with EduRAG (CLAG assesses level) → Receive CAQ quiz → Rate appropriateness
- Metrics: Subjective appropriateness (target: > 3.8/5), Quiz accuracy (target: 60–80%)

4. System Architecture

4.1 Knowledge Grounding Infrastructure

For knowledge grounding, we adopt the Structure-Aware RAG paradigm established by KiRAG (Fang et al., 2025) and RECOMP (Xu et al., 2023):

Knowledge Unit Extraction. Given retrieved chunks $\mathcal{D} = \{d_1, \dots, d_k\}$:

$$\mathcal{KU} = \text{Extract}(\mathcal{D}) = \{ku_1, ku_2, \dots, ku_m\}$$

Each $ku_i = (\text{content}_i, \text{source}_i, \text{conf}_i)$.

We implement T5-based extraction (following RECOMP) and extend with **educational knowledge typing**: course-specific (requires citation), general CS knowledge (acceptable without citation), common knowledge (no verification needed).

Constrained Generation. Soft-constraint prompting following FAIR-RAG (Asl et al., 2025), with generation-time citation tagging (the G-Cite paradigm; Saxena et al., 2025).

4.2 Pipeline Orchestration

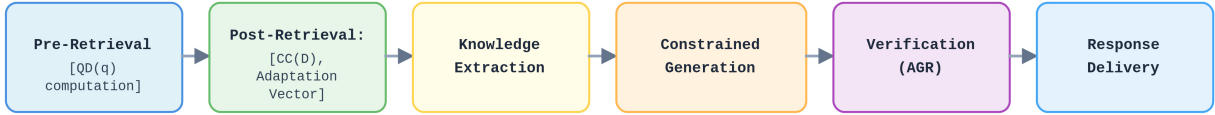


Figure 1: Pipeline

Query complexity routing follows Adaptive-RAG (Jeong et al., 2024). Retrieval verification uses single-retry gates inspired by CRAG (Yan et al., 2024). No novelty claimed in orchestration.

4.3 Technology Stack

Component	Technology	Rationale
LLM	LLaMA 3.1 8B (4-bit GGUF)	Offline capability, open weights
Inference	llama.cpp	Optimized CPU/GPU inference
Vector DB	ChromaDB	Lightweight, embedded
Embeddings	all-MiniLM-L6-v2	384 dims, fast, good quality
Frontend	Electron + React	Cross-platform desktop
Fine-tuning	QLoRA	Memory-efficient adaptation

4.4 Deployment Target

- **Platform:** Offline desktop application
- **Hardware:** Consumer laptop (16GB RAM, optional GPU)
- **Latency target:** < 3 seconds end-to-end
- **Storage:** ~8GB (model + index)

5. Dataset Contributions

5.1 Fine-Tuning Corpus

Characteristic	Specification
Size	2,000+ instruction-response pairs
Source	CS curriculum (semesters 2–8)

Characteristic	Specification
Format	Alpaca-style (instruction, input, output)
Bloom’s Distribution	Stratified across 6 levels
Quality Control	Human review of 10% sample

Categories: Conceptual explanations (35%), Diagram description (25%), Code generation (20%), Problem solving (10%), Quiz generation (10%).

5.2 Evaluation Benchmark

Characteristic	Specification
Size	2,000+ queries with ground truth
Stratification	6 Bloom’s levels $\times$ 8 CS topics
Annotations	Correct answer, relevant sources, difficulty
IAA	Target $\kappa > 0.7$

5.3 Release Plan

- **Format:** HuggingFace Datasets + Zenodo archive
- **License:** CC-BY-4.0
- **Documentation:** Datasheet following Gebru et al. (2021)

6. End-to-End Experimental Plan

6.1 System-Level Baselines

Baseline	Description
Vanilla RAG	Retrieve + Generate, no KU extraction, no adaptation
KiRAG-style	KU extraction + constrained generation, no CLAG/AGR
EduRAG-Fixed	Full pipeline with fixed Medium adaptation level
EduRAG-Full	Full pipeline with CLAG + AGR

6.2 End-to-End Metrics

Metric	Target	Measurement
Latency	$< 3s$ per query	Automated timing
Hallucination Rate	$< 10\%$	Expert annotation
Citation Accuracy	$> 90\%$	Expert annotation ($n = 200$)
User Satisfaction	$> 4.0/5.0$	Likert survey ($N = 30$)
Learning Gain	$> 20\%$	Pre/post test ($N = 60$)

7. Timeline

Phase	Duration	Activities
Phase 1	Weeks 1–2	Corpus collection, preprocessing, KGV infrastructure
Phase 2	Weeks 3–4	ERSA experiments (44 conditions), CLAG v2 implementation
Phase 3	Weeks 5–6	User studies (CLAG, PCO), AGR benchmark + evaluation
Phase 4	Weeks 7–8	Integration, CAQ, end-to-end testing
Phase 5	Weeks 9–12	Paper writing, dataset release

9. Expected Outcomes

1. **Validated framework** for stateless cognitive adaptation (CLAG), demonstrating the Memory Dogma can be relaxed
2. **New verification paradigm** (AGR) addressing the adaptation-grounding tension with derivation rules
3. **Empirical scaling model** (ERSA/RSC) for educational RAG capacity planning
4. **Open datasets** enabling reproducible educational RAG research
5. **Working system** demonstrating offline educational AI feasibility on consumer hardware

10. Limitations and Scope

Limitation	Mitigation
CS domain only	Architecture generalizes; domain-specific tuning required
User study sample size ($N = 60$)	Sufficient for medium effect detection; acknowledged
QD correction cap $\delta_q = 0.10$ is a tunable hyperparameter	Theoretically derived (same order as σ_B); sensitivity analysis in Experiment C1
AGR derivation rules are enumerated, not learned	Immediate future work: learned derivation engine
Offline deployment constraints	Feature, not limitation, addresses accessibility gap

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